01 · LTspice from zero

Needed for Exps 2, 4, 5, 8, 9, 10, 11, 12 and the extra experiment · LTspice is free (Analog Devices) and already installed on your laptop

Where this sits

LTspice basics → the 4 analyses → drawing a circuit → reading the result → exporting data

What you already have: circuits from C&S (KVL, KCL, sources, R, C), and the idea that a graph of one quantity against another describes a device.

Three terms first:

- **Simulator:** a program that solves the circuit's KVL/KCL equations for you, using a mathematical **model** of each part.
- **Model:** the set of numbers that describes a real part. For example, "BC547B" tells LTspice the β, the junction capacitances, and so on.
- **Directive:** a text command starting with a dot (`.dc`, `.ac`, `.tran`, `.op`, `.step`). It tells LTspice *which* analysis to run.

MANUAL EXP 2, FIG 2.1

The circuit has `.dc V1 0 20 1` written under it. What does that line make LTspice do?

GUESS FIRST

Guess what each of the four words means: `.dc`, `V1`, `0 20`, `1`.

The four analyses (every software experiment uses one or more)

Directive	What it does	Gives you	Used in
`.op`	finds the DC operating point (all capacitors open)	one number per node	Exp 2 (load), checking a Q-point
`.dc V1 0 20 1`	sweeps source V1 from 0 to 20 V in 1 V steps, solving at each step	a curve: anything against V1	Exps 2, 5, 9
`.ac dec 10 200 1Meg`	small-signal sine at each frequency, 10 points per decade, 200 Hz to 1 MHz	gain (dB) and phase against frequency	Exps 7, 8, 10, extra
`.tran 10m`	runs time forward for 10 ms	waveforms against time	Exps 4, 7, 10, 11, 12
`.step param R 100 1100 100`	reruns the analysis for R = 100, 200, … 1100	a family of curves	Exp 2 (load)

So `.dc V1 0 20 1` = "DC sweep of V1, start 0, stop 20, step 1". A **nested** sweep has a second source at the end: `.dc Vce 0 12 0.05 Ib 0 60u 20u` sweeps V_{ce} for each of I_b = 0, 20, 40, 60 µA, which gives one curve per I_b.

Units: the trap that costs marks

SPICE ignores case, so `M` **means milli, not mega.**

You want	Type	Not
1 MΩ	`1Meg`	~~`1M`~~ (that is 1 mΩ)
10 µF	`10u`	
0.5 nF	`0.5n` or `500p`	
2.2 kΩ	`2.2k`	

Drawing a circuit

DO IT: DRAW, SET, RUN

1. **File → New Schematic.**
2. Place parts using the keyboard: **R** resistor, **C** capacitor, **D** diode, **G** ground. **F2** opens the full component list (`voltage`, `npn`, `njf`, `nmos`, `zener`).
3. **F3** draws a wire: click at the start, click at each corner, right-click to stop. **Ctrl+R** rotates a part before you place it.
4. **Every circuit needs a ground (G).** Without one, the run fails with a "singular matrix" or "floating node" error.
5. **Right-click a part** to set its value (e.g. `2.2k`).
6. To choose a **model** (BC547B, 2N3819, 1N914, BZX84C6V2L, FDR6580, 2N2222, 2N5434), right-click the diode or transistor and click **Pick New ...**. All the manual's parts are in LTspice's library.
7. To set up a **source**, right-click it and click **Advanced**: - **SINE**: DC offset, amplitude, frequency. `SINE(0 5 1k)` = 0 V offset, 5 V peak, 1 kHz. - **Small signal AC amplitude**: type `1` (or `0.5`) here for an `.ac` run. - **PULSE**: V1, V2, delay, rise, fall, on-time, period, cycles.
8. To add a directive, press **S** (or **Edit → SPICE Directive**), type it, and place the text anywhere on the sheet. Alternatively, **Simulate → Edit Simulation Cmd** fills the directive in for you.
9. **F4** puts a label on a wire. Name the output `Vo` so it is easy to plot.
10. Click **Run** (the running-man icon).

Reading the result

You want	Do this
a node voltage	after the run, click on the wire (the probe turns into a red voltage probe)
a current	click on the part's body (the probe turns into a current clamp)
a different x-axis (e.g. V(base) instead of V_{be})	right-click the x-axis, then set **Quantity plotted** to `V(base)`
an exact value	click the trace's name above the plot. A cursor appears, and its readout box shows x and y. A second click adds cursor 2; the box then also shows the difference.
gain in plain ratio, not dB	right-click the y-axis and choose **Linear**
y-axis limits	right-click the y-axis and set **Manual limits**

In an `.ac` plot, the **solid line is the gain in dB** and the **dashed line is the phase**.

OBSERVATION TABLES FROM A SIMULATION

The manual says: "Paste data points from simulations". With the plot window active, use **File → Export data as text**. The file has one row per point. Copy only the rows the table asks for (5–10 points), and mark on the graph the points used in calculations.

TRAP: THE DEFAULT TIME STEP

`.tran 20m` lets LTspice pick its own step, which can be coarse. Edges then look slanted, or tall false spikes appear. Write `.tran 0 20m 0 1u` instead: the fourth number caps the step at 1 µs.

CHECK 1

(a) Write the directive that sweeps V_{gs} from −5 V to 0 V in 10 mV steps. (b) Your 1 MΩ resistor shows `1M`. What value is LTspice actually using? (c) Which analysis gives a frequency response?

Answers

Check 1. (a) `.dc Vgs -5 0 0.01` (or `10m`). (b) 1 milliohm, a short circuit. Write `1Meg`. (c) `.ac`, with the source's AC amplitude set.

Opening question. `.dc` = DC sweep, `V1` = the source being swept, `0 20` = from 0 V to 20 V, `1` = in 1 V steps. LTspice solves the circuit 21 times and plots any node against V1.